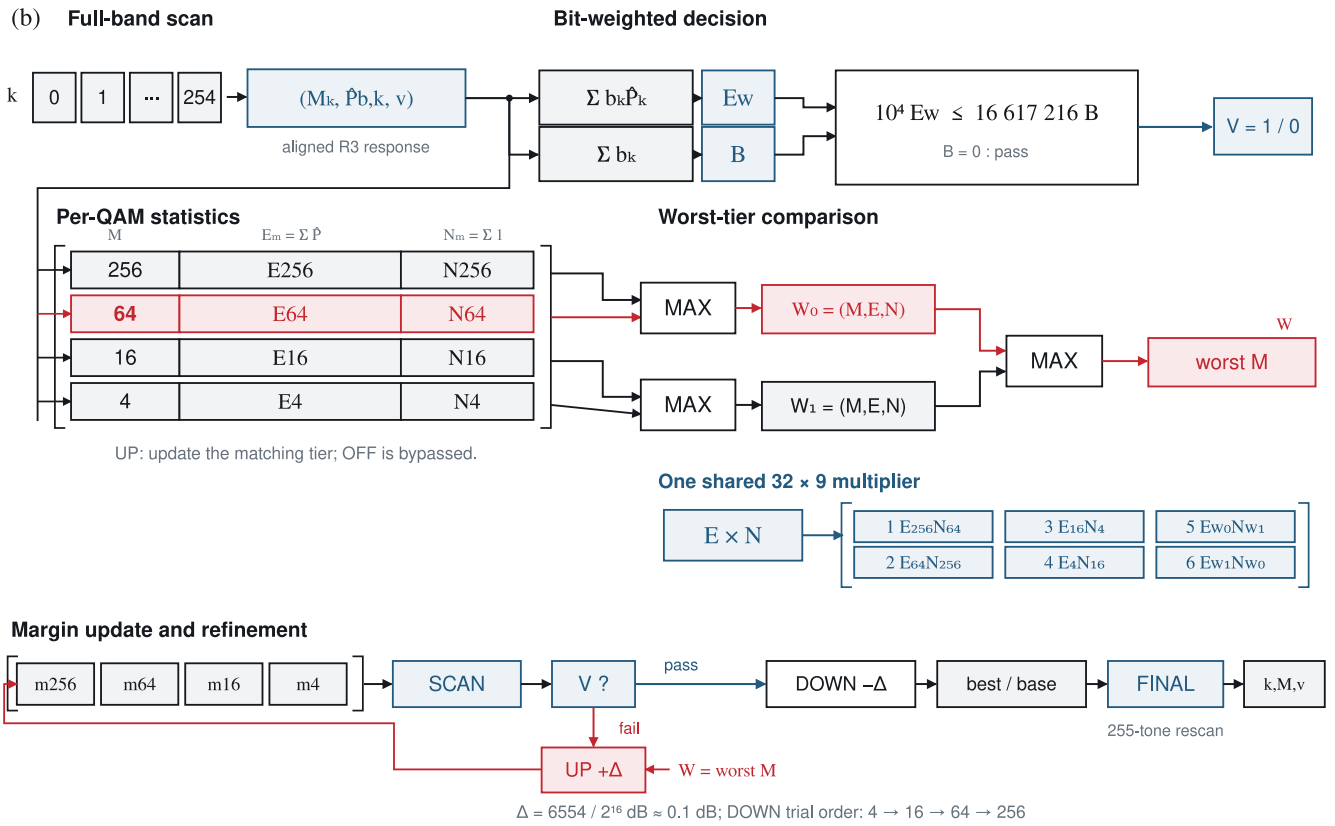

(b) Full-band scan
Bit-weighted decision
k
0
1
...
254
(Mk, P̂b,k, v)
aligned R3 response
Σ bkP̂k
Σ bk
Ew
B
10⁴ Ew ≤ 16 617 216 B
B = 0 : pass
V = 1 / 0
Per-QAM statistics
M
Em = Σ P̂
Nm = Σ 1
256
E256
N256
64
E64
N64
16
E16
N16
4
E4
N4
UP: update the matching tier; OFF is bypassed.
Worst-tier comparison
MAX
W0 = (M,E,N)
MAX
W1 = (M,E,N)
MAX
W
worst M
One shared 32 × 9 multiplier
E × N
1 E256N64
3 E16N4
5 Ew0Nw1
2 E64N256
4 E4N16
6 Ew1Nw0
Margin update and refinement
m256
m64
m16
m4
SCAN
V ?
pass
DOWN −Δ
best / base
FINAL
k,M,v
255-tone rescan
fail
UP +Δ
W = worst M
Δ = 6554 / 2¹⁶ dB ≈ 0.1 dB; DOWN trial order: 4 → 16 → 64 → 256